Alpha Spectrometry Standards

Eckert & Ziegler

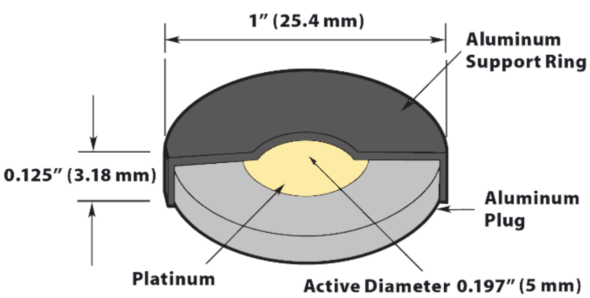
Alpha spectrometry standards are intended for calibrating and checking solid state alpha spectrometers for a wide range of applications. Most radionuclides are plated onto a stainless steel or platinum disk by either electrodeposition or electro-less deposition and annealed. Some uranium sources are electrodeposited onto aluminum disks and Po-210 is plated onto silver disks. These processes produce sources exhibiting peak widths typically less than 20 keV full width half maximum (sources containing Np-237 and U-238 have peak widths typically 50 keV FWHM, due to their low specific activity). Alpha spectrometry standards are available for the alpha emitting radionuclide listed on the second page. Standards are offered with either a single radionuclide or a mixture of these radionuclides.

Calibration

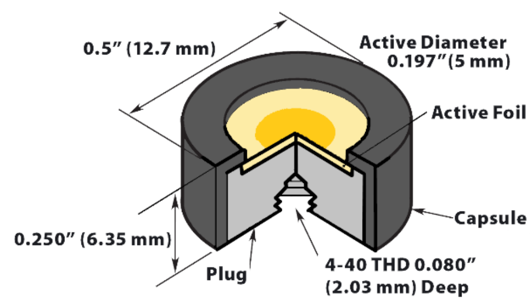
Alpha spectrometry standards are manufactured in our three ISO 17025 accredited calibration laboratories. Standards can be calibrated for contained activity and/or alpha emission rate. Alpha spectrometry sources can also be provided without calibration for measurements of alpha energy only.

Plated onto Platinum

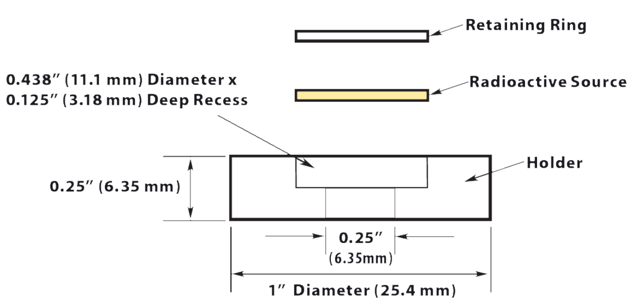
Alpha spectrometry standards plated onto platinum disks are mounted in the Type A1, Type A2 or Type PM capsules and are available with a gold cover up to 200 µg/cm^2 upon request. For Ra-226 and Th-228 standards, a 100 µg/cm^2 gold cover is recommended to prevent loss of radioactive recoil daughter products. A 100 µg/cm^2 gold cover is also recommended for Cf-252 sources, however the gold will not completely prevent the loss of fission fragments. Po-210 standards plated onto silver and uranium standards plated onto aluminum are also available in the Type A1, Type A2 or Type PM capsules and can be made with a 100 µg/cm^2 acrylic cover upon request.



Type A1 sources are permanently fixed in an aluminum holder 1” diameter x 0.125” high (25.4 mm x 3.18 mm). The active diameter is 0.197” (5.0 mm).



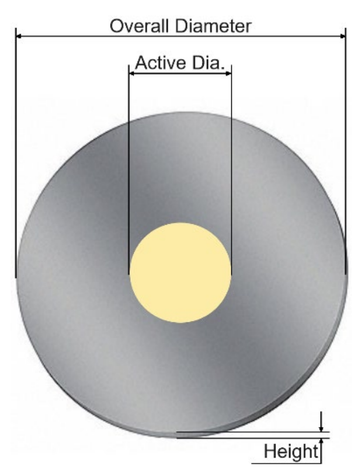
Type A2 source are permanently fixed in an aluminum holder 0.5” diameter x 0.250” high (12.7 mm x 6.35 mm). The active diameter is 0.197” (5.0 mm).



Type PM sources are mounted in a plastic holder from which it can be separated for installation in a counting chamber or device. The holder is 1” diameter x 0.125” high (25.4 mm x 3.18 mm). The removable active foil is 0.438” (11.1 mm) in diameter with the active diameter 0.197” (5.0 mm). The foils are platinum between 0.005” and 0.010” (0.127 mm and 0.254 mm) thick.

Plated onto Stainless Steel

Alpha spectrometry standards plated onto stainless steel are available on either a Ø 25 mm x 0.5 mm disk with 7 mm active diameter or a Ø 24.1 mm x 0.77 mm disk with any specified active diameter from 5 mm to 22 mm. Alpha spectrometry standards plated on stainless steel are only available uncovered.



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Nuclide	Half-life	Alpha Energies (keV)	Platinum Disks	Stainless Steel Disks	Other Disks
Am-241	432.2 y	5388, 5443, 5486	A1, A2, PM Capsules	Ø 25 mm x 0.5 mm Ø 24.1 mm x 0.77 mm	
Am-243	7.364×10^3 y	5181, 5233, 5275		Ø 24.1 mm x 0.77 mm	
Cf-252	2.645 y	6076, 6118	A1, A2, PM Capsules		
Cm-244	18.11 y	5763, 5805	A1, A2, PM Capsules	Ø 25 mm x 0.5 mm Ø 24.1 mm x 0.77 mm	
Gd-148	75 y	3184	A1, A2, PM Capsules	Ø 24.1 mm x 0.77 mm	
Np-237	2.14×10^6 y	4640-4873	A1, A2, PM Capsules	Ø 25 mm x 0.5 mm Ø 24.1 mm x 0.77 mm	
Po-210	138.376 d	5304			Silver Foil in A1, A2, PM Capsules
Pu-238	87.74 y	5456, 5499	A1, A2, PM Capsules	Ø 25 mm x 0.5 mm Ø 24.1 mm x 0.77 mm	
Pu-239	2.411×10^4 y	5105, 5143, 5156	A1, A2, PM Capsules	Ø 25 mm x 0.5 mm Ø 24.1 mm x 0.77 mm	
Ra-226	1,600 y	4601, 4784	A1, A2, PM Capsules		
Th-228	698.2 d	5341, 5423	A1, A2, PM Capsules	Ø 24.1 mm x 0.77 mm	
Th-230	7.54×10^4 y	4621, 4688	A1, A2, PM Capsules	Ø 25 mm x 0.5 mm Ø 24.1 mm x 0.77 mm	
U-233	1.592×10^5 y	4729, 4784, 4824		Ø 24.1 mm x 0.77 mm	
U-234	2.454×10^5 y	4722, 4775		Ø 24.1 mm x 0.77 mm	
U-235	7.037×10^8 y	4215-4597		Ø 24.1 mm x 0.77 mm	Aluminum Foil in A1, A2, PM Capsules
U-236	2.342×10^7 y	4445, 4494		Ø 24.1 mm x 0.77 mm	
U-238(Nat)	4.468×10^9 y	4147, 4196		Ø 24.1 mm x 0.77 mm	Aluminum Foil in A1, A2, PM Capsules
U-238(Dep)	4.468×10^9 y	4147, 4196		Ø 24.1 mm x 0.77 mm	Aluminum Foil in A1, A2, PM Capsules

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